Revision

REVISION

Document Identification : FMCOMDOCMODBUSV0100

Revision No	Changes	Date
0	New Document	03/04/2018

1 LED Status Address has been added. Current sensor and Voltage Sensor Address is added

2 Fault record Trip elements are corrected and Event record Trip messages are added.

3 CBF setting ranges are added in protection settings. Hot Line tag is added in event record J Register. Corrected spelling mistakes in protection settings G9 register. Corrected register size in write command.

Correct the Multiplication Factor for Fault record

4 and Measurand.

MD Apparent Power Reverse address, Frequency

5 multiplication factor has been corrected in Measurand. Add Control Changed event and Breaker Close name changed to Breaker status in Event record. Spontaneous Events address is added. Realtime

6 Rly Function address is added in System settings, Output relay and LED settings. Address modified for DI Timer & Blocking, LED status, Sensor value.

Reg_25024: Added negative sign for T.KW-

7 Export & positive sign for T.KW-Import.

Reg_25026: Added negative sign for T.KVAR-Lg & positive sign for T.KVAR-Ld.

Reg_25028: Added T.KVA value.

Contents

TABLE OF CONTENTS

1 Communication Parameters 4

2 Measurements 4

3 Settings 6

3.1 System Settings ……

3.2 Protection Settings ……

3.3 Auto Reclose Settings ……

3.4 Cold Load Settings ……

3.5 Demand Control Settings ……

3.6 CB Maintaince Settings ……

3.7 Sectionalizer Settings ……

3.8 Output Relay Settings ……

3.9 LED Settings ……

3.10 Digital Input Timer Settings **17**

3.11 Digital Input Blocking Settings ……..…**20**

3.12 LED Status ……..**23**

4 Commands **23**

5 Time Synchronization **23**

6 Fault Record **24**

7 Event Record **26**

8 Events **27**

---

1 Communication Parameters

MODBUS MEMORY MAPPING

1 COMMUNICATION PARAMETERS:

The communication between the Master and Slave (Relay) is established using the following paran when given correctly establish the communication.

Sl No	Parameters	Range
1	Station Address/Device Id	1 to 247
2	Baud Rate	1200 , 2400 , 4800 , 9600 , 19200
3	Stop Bits	1,2
4	Parity	None , Odd , Even
5	Data Bits	7 , 8

2 MEASUREMENTS:

Address	Register Name	Range	Register Size	Step	Multip Factor
40001	R Phase Protection Current	-	4 Bytes	-	0.
40003	Y Phase Protection Current	-	4 Bytes	-	0.

40005	B Phase Protection Current	-	4 Bytes	-	0
40007	Earth Current	-	4 Bytes	-	0
40009	Sensitive Earth Fault Current	-	4 Bytes	-	0
40011	I1 Current-Positive Sequence	-	4 Bytes	-	0
40013	I2 Current-Negative Sequence	-	4 Bytes	-	0
40015	I0 Current-Zero Sequence	-	4 Bytes	-	0
40017	R Phase Meter Current	-	4 Bytes	-	0
40019	Y Phase Meter Current	-	4 Bytes	-	0
40021	B Phase Meter Current	-	4 Bytes	-	0
40023	R Phase L-N Voltage	-	4 Bytes	-	0.00
40025	Y Phase L-N Voltage	-	4 Bytes	-	0.00
40027	B Phase L-N Voltage	-	4 Bytes	-	0.00
40029	RY Line-Line Voltage	-	4 Bytes	-	0.00
40031	YB Line-Line Voltage	-	4 Bytes	-	0.00
40033	BR Line-Line Voltage	-	4 Bytes	-	0.00
40035	Residual Voltage	-	4 Bytes	-	0.00
40037	V1 Voltage-Positive Sequence	-	4 Bytes	-	0.00
40039	V2 Voltage-Negative Sequence	-	4 Bytes	-	0.00
40041	V0 Voltage-Zero Sequence	-	4 Bytes	-	0.00
40043	R Phase L-N Output Voltage	-	4 Bytes	-	0.00
40045	Y Phase L-N Output Voltage	-	4 Bytes	-	0.00
40047	B Phase L-N Output Voltage	-	4 Bytes	-	0.00
40049	R Phase Reverse KW	-	4 Bytes	-	0
40051	R Phase Forward KW	-	4 Bytes	-	0
40053	Y Phase Reverse KW	-	4 Bytes	-	0
40055	Y Phase Forward KW	-	4 Bytes	-	0.001
40057	B Phase Reverse KW	-	4 Bytes	-	0.001
40059	B Phase Forward KW	-	4 Bytes	-	0.001
40061	R Phase Reverse KVA	-	4 Bytes	-	0.001
40063	R Phase Forward KVA	-	4 Bytes	-	0.001
40065	Y Phase Reverse KVA	-	4 Bytes	-	0.001
40067	Y Phase Forward KVA	-	4 Bytes	-	0.001
40069	B Phase Reverse KVA	-	4 Bytes	-	0.001
40071	B Phase Forward KVA	-	4 Bytes	-	0.001
40073	R Phase Power Factor	-	4 Bytes	-	0.001
40075	Y Phase Power Factor	-	4 Bytes	-	0.001
40077	B Phase Power Factor	-	4 Bytes	-	0.001
40079	R Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40081	R Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40083	R Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40085	R Phase KVAR Forward Lead	-	4 Bytes	-	0.001
40087	Y Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40089	Y Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40091	Y Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40093	Y Phase KVAR Forward Lead	-	4 Bytes	-	0.001

40095	B Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40097	B Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40099	B Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40101	B phase KVAR Forward Lead	-	4 Bytes	-	0.001
40103	Total KW Reverse	-	4Bytes	-	0.001
40105	Total KW Forward	-	4 Bytes	-	0.001
40107	Total KVA Reverse	-	4 Bytes	-	0.001
40109	Total KVA Forward	-	4 Bytes	-	0.001
40111	Total KVAR Reverse Lag	-	4 Bytes	-	0.001
40113	Total KVAR Reverse Lead	-	4 Bytes	-	0.001
40115	Total KVAR Forward Lag	-	4 Bytes	-	0.001
40117	Total KVAR Forward Lead	-	4 Bytes	-	0.001
40119	R Active Energy Reverse	-	4 Bytes	-	0.001
40121	Y Active Energy Reverse	-	4 Bytes	-	0.001
40123	B Active Energy Reverse	-	4 Bytes	-	0.001
40125	R Active Energy Forward	-	4 Bytes	-	0.001
40127	Y Active Energy Forward	-	4 Bytes	-	0.001
40129	B Active Energy Forward	-	4 Bytes	-	0.001
40131	R Apparent Energy Reverse	-	4 Bytes	-	0.001
40133	Y Apparent Energy Reverse	-	4 Bytes	-	0.001
40135	B Apparent Energy Reverse	-	4 Bytes	-	0.001
40137	R Apparent Energy Forward	-	4 Bytes	-	0.001
40139	Y Apparent Energy Forward	-	4 Bytes	-	0.001
40141	B Apparent Energy Forward	-	4 Bytes	-	0.001
40143	R KVARH Reverse Lead	-	4 Bytes	-	0.001
40145	Y KVARH Reverse Lead	-	4 Bytes	-	0.001
40147	B KVARH Reverse Lead	-	4 Bytes	-	0.001
40149	R KVARH Reverse Lag	-	4 Bytes	-	0.001
40151	Y KVARH Reverse Lag	-	4 Bytes	-	0.001
40153	B KVARH Reverse Lag	-	4 Bytes	-	0.001
40155	R KVARH Forward Lead	-	4 Bytes	-	0.001
40157	Y KVARH Forward Lead	-	4 Bytes	-	0.001
40159	B KVARH Forward Lead	-	4 Bytes	-	0.001
40161	R KVARH Forward Lag	-	4 Bytes	-	0.001
40163	Y KVARH Forward Lag	-	4 Bytes	-	0.001
40165	B KVARH Forward Lag	-	4 Bytes	-	0.001
40167	Total Active Energy Reverse	-	4 Bytes	-	0.001
40169	Total Active Energy Forward	-	4 Bytes	-	0.001
40171	Total Apparent Energy Reverse	-	4 Bytes	-	0.001
40173	Total Apparent Energy Forward	-	4 Bytes	-	0.001
40175	Total KVARH Reverse Lead	-	4 Bytes	-	0.001
40177	Total KVARH Reverse Lag	-	4 Bytes	-	0.001
40179	Total KVARH Forward Lead	-	4 Bytes	-	0.001
40181	Total KVARH Forward Lag	-	4 Bytes	-	0.001
40183	Average Power Factor	-	4 Bytes	-	0.001

40185	MD Apparent Power Reverse	-	4 Bytes	-	0.001
40187	MD Apparent Power Forward	-	4 Bytes	-	0.001
40189	MD Active Power Reverse	-	4 Bytes	-	0.001
40191	MD Active Power Forward	-	4 Bytes	-	0.001
40193	Cumulative MD KVA Reverse	-	4 Bytes	-	0.001
40195	Cumulative MD KVA Forward	-	4 Bytes	-	0.001
40197	Cumulative MD KW Reverse	-	4 Bytes	-	0.001
40199	Cumulative MD KW Forward	-	4 Bytes	-	0.001
40201	Frequency	-	4 Bytes	-	0.01
40203	Last Outage Bus Days-2 Bytes,Hours,Minutes	-	4Bytes	-	1
40205	Total Outage Bus Days-2 Bytes,Hours,Minutes	-	4 Bytes	-	1
40207	Last Outage Load Days-2 Bytes,Hours,Minutes	-	4Bytes	-	1
40209	Total Outage Load Days-2 Bytes,Hours,Minutes	-	4 Bytes	-	1
40211	Outage Count Bus	-	4 Bytes	-	1
40213	Outage Count Load	-	4 Bytes	-	1
40215	Auto Trip Count	-	4 Bytes	-	1
40217	Sum Of I^2 Alarm	-	4 Bytes	-	1
25024	3Ph Total KW	-	4 Bytes	-	0.001
25026	3Ph Total KVAR	-	4 Bytes	-	0.001
25028	3Ph Total KVA	-	4 Bytes	-	0.001

3 SETTINGS:

3.1 System Settings:

Address	Register Name	Range	Register Size	Step	Multiplicati Factor
40284-	Identifier	-	16 Bytes	-	1
40299					
40300	Current Sensor Direction	Fwd-1,Rev-0	1 Byte	1	1
40301	No Volt UV Block	Yes-1,No-0	1 Byte	1	1
40302	Cold Load	ON-1,OFF-0	1 Byte	1	1
40303	Auto Reclose	ON-1,OFF-0	1 Byte	1	1
40304	Sectionalizer Setting	ON-1,OFF-0	1 Byte	1	1
40305	Breaker Charging	OFF-0, DeadBus-1, DeadLine-2	1 Byte	1	1
40306	OV Operate Mode	Single Phase-1, Three Phase-0	1 Byte	1	1
40307	UV Operate Mode	Single Phase-1, Three Phase-0	1 Byte	1	1
40308	Active Group	G1-GT(1-10)	1 Byte	1	1
40309	Edit Group	G1-GT(1-10)	1 Byte	1	1
40310	Digital Input1 Configuration	*Note 1	1 Byte	1	1

40311	Digital Input2 Configuration	*Note 1	1 Byte	1	1
40312	Digital Input3 Configuration	*Note 1	1 Byte	1	1
40313	Digital Input4 Configuration	*Note 1	1 Byte	1	1
40314	Digital Input5 Configuration	*Note 1	1 Byte	1	1
40315	Digital Input6 Configuration	*Note 1	1 Byte	1	1
40316	Digital Input7 Configuration	*Note 1	1 Byte	1	1
40317	Digital Input8 Configuration	*Note 1	1 Byte	1	1
40318	Digital Input9 Configuration	*Note 1	1 Byte	1	1
40319	Digital Input10 Configuration	*Note 1	1 Byte	1	1
40320	Digital Input11 Configuration	*Note 1	1 Byte	1	1
40321	Digital Input12 Configuration	*Note 1	1 Byte	1	1
40322	Digital Input13 Configuration	*Note 1	1 Byte	1	1
40323	Digital Input14 Configuration	*Note 1	1 Byte	1	1
40324	Digital Input15 Configuration	*Note 1	1 Byte	1	1
40325	Push Button 1	*Note 2	1 Byte	1	1
40326	Push Button 2	*Note 2	1 Byte	1	1
41142	Current Sensor	1-3	1 Byte	1	1
	Voltage Sensor	1-5	1 Byte	1	1
41143	Real Time RLY1	ON-1,OFF-0	1 Byte	1	1
41144	Real Time RLY2	ON-1,OFF-0	1 Byte	1	1
41145	Real Time RLY1 On Hr	00-23	1 Byte	1	1
41146	Real Time RLY1 On Min	00-59	1 Byte	1	1
41147	Real Time RLY1 On Sec	00-59	1 Byte	1	1
41148	Real Time RLY1 Off Hr	00-23	1 Byte	1	1
41149	Real Time RLY1 Off Min	00-59	1 Byte	1	1
41150	Real Time RLY1 Off Sec	00-59	1 Byte	1	1
41151	Real Time RLY2 On Hr	00-23	1 Byte	1	1
41152	Real Time RLY2 On Min	00-59	1 Byte	1	1
41153	Real Time RLY2 On Sec	00-59	1 Byte	1	1
41154	Real Time RLY2 Off Hr	00-23	1 Byte	1	1
41155	Real Time RLY2 Off Min	00-59	1 Byte	1	1
41156	Real Time RLY2 Off Sec	00-59	1 Byte	1	1

Note 1:
0 – OFF
1 – ON
2 – ON, Group Change
4 – ON, Reset Device
5 – ON, Remote Close
6 – ON, Remote Trip S
7 – ON, Remote Trip H
Note 2:
0 – OFF
1 – ON
2 – ON, Group Change
4 – ON, Reset Device
5 – ON, Close
6 – ON, Trip S
7 – ON, Trip H
8 – ON, Local/Remote
3.2 Protection Settings:

Address	Register Name	Range	Register Size	Step	Multiplicat Factor
40327	Phase Current (A)	Depends upon the Sensor Rating	2 Bytes	1	1
40328	Phase Characteristics	1-6	1 Byte	1	1
40329	Phase Time Multiplier (S)	1– 1600	2 Bytes	1	0.001
40330	Phase Definite Time Lag(S)	0 – 20000	2 Bytes	1	0.01

40331	Phase Wideset (A)	Depends upon the Sensor Rating	2 Bytes	1	1
40332	Phase Wideset Time (S)	0 – 20000	2 Bytes	1	0.001
40333	Phase Highset (A)	Depends upon the Sensor Rating	2 Bytes	1	1
40334	Earth Current (A)	Depends upon the	2 Bytes	1	1
		Sensor Rating			
40335	Earth Characteristics	1-6	1 Byte	1	1
40336	Earth Time Multiplier (S)	1-1600	2 Bytes	1	0.001
40337	Earth Definite Time Lag(S)	0-20000	2 Bytes	1	0.01
40338	Earth Wideset (A)	Depends upon the Sensor Rating	2 Bytes	1	1
40339	Earth Wideset Time (S)	0-20000	2 Bytes	1	0.001
40340	Earth Highset (A)	Depends upon the Sensor Rating	2 Bytes	1	1
40341	Reset Time (S)	1-60	2 Bytes	1	1
40342	Circuit Breaker Failure Set(A)	Depends upon the Sensor Rating	2 Bytes	1	1
40343	Circuit Breaker Failure Delay (S)	0-100	2 Bytes	1	0.01
40344	Sensitive Earth Fault Set(A)	Depends upon the Sensor Rating	2 Bytes	1	1
40345	Sensitive Earth Fault Time(S)	0-2000	2 Bytes	1	0.01
40346	Over Voltage Stage1 (V)	Depends upon the Sensor Rating	2 Bytes	1	1
40347	Over Voltage Stage1 Characteristics	1-2	1 Bytes	1	1
40348	Over Voltage Time Multiplier (S)	25-10000	2 Bytes	1	0.01
40349	Over Voltage Definite Time Lag (S)	0-60000	2 Bytes	1	0.01
40350	Over Voltage Stage2 (V)	Depends upon the Sensor Rating	2 Bytes	1	1
40351	Over Voltage Stage2 Definite Time Lag (S)	0-60000	2 Bytes	1	0.01
40352	Over Voltage Reset Time(S)	1-60	2 Bytes	1	1
40353	Under Voltage Stage1 (V)	Depends upon the Sensor Rating	2 Bytes	1	1
40354	Under Voltage Stage1 Characteristics	1-6	1 Bytes	1	1
40355	Under Voltage Stage1 Time Multiplier (S)	25-10000	2 bytes	1	0.01
40356	Under Voltage Stage1 Definite Time Lag (S)	0-60000	2 bytes	1	0.01
40357	Under Voltage Stage 2(V)	Depends upon the Sensor Rating	2 Bytes	1	1

40358	Under Voltage Stage2 Definite Time Lag (S)	0-60000	2 Bytes	1	0.01
40359	Under Voltage Reset Time(S)	1-60	2 Bytes	1	1
40360	Residual Voltage Stage1(V)	Depends upon the Sensor Rating	2 Bytes	1	1
40361	Residual Voltage Characteristics	1-2	2 Bytes	1	1
40362	Residual Voltage Time Multiplier (S)	25-10000	2 Bytes	1	0.01
40363	Residual Voltage Definite Time Lag (S)	0-60000	2 Bytes	1	0.01
40364	Residual Voltage Stage2 (V)	Depends upon the Sensor Rating	2 Bytes	1	1
40365	Residual Voltage Stage2 Definite Time Lag (S)	0-60000	2 Bytes	1	0.01
40366	Residual Voltage Reset Time (S)	1-60	2 Bytes	1	1

Sensor Rating	**Phase Set**	**Phase Wideset / Highset**	**Earth Set**	**Earth Wideset / Highset**	**CB Fail Set**
60 A	3-120 A	3-1800 A	3-120 A	3-1800 A	3-120 A
250 A	10-500 A	10-7500 A	10-500 A	10-7500 A	10-500 A
750 A	30-1500 A	30-22500 A	30-1500 A	30-22500 A	30-1500 A

Sensor Rating	**OV Stage1**	**OV Stage2**	**UV Stage1**	**UV Stage2**	**RES Stage1**
1.905 KV	1500-3240V	1500-3240V	150-2280V	150-2280V	20-600V
3.811 KV	3000-6480V	3000-6480V	300-4560V	300-4560V	40-1200V
6.351 KV	5000-10800V	5000-10800V	500-7600V	500-7600V	60-2000V
12.702 KV	10000-21600V	10000-21600V	1000-15200V	1000-15200V	120-4000V
19.053 KV	15000-32400V	15000-32400V	1500-22800V	1500-22800V	180-6000V

The above address values are for Group1 Settings of the Protection settings the Protection settings informatior same as that of the Group1 except their address are different from Group1. The settings are of same order for a and ending address of the other Group setting are given below

Group	**Starting Address**	**Ending Address**
Protection Settings Group 2	40367	40406
Protection Settings Group 3	40407	40446
Protection Settings Group 4	40447	40486
Protection Settings Group 5	40487	40526
Protection Settings Group 6	40527	40566
Protection Settings Group 7	40567	40606
Protection Settings Group 8	40607	40646

Protection Settings Group 9	40647	40686
Protection Settings Group 10	40687	40726

3.3 Auto Reclose Setting:

Address	Register Name	Range	Register Size	Step	Multiplicati Factor
40727	Auto Shots	1-4	2 Bytes	1	1
40728	Reclaim Time (M)	5-6000	2 Bytes	1	0.01
40729	LockOut H.Reset Time (S)	50-60000	2 Bytes	1	0.01
40730	Phase Fault Reclose Time 1 (S)	50-60000	2 Bytes	1	0.01
40731	Phase Fault Reclose Time 2 (S)	50-60000	2 Bytes	1	0.01
40732	Phase Fault Reclose Time 3 (S)	50-60000	2 Bytes	1	0.01
40733	Phase Fault Reclose Time 4 (S)	50-60000	2 Bytes	1	0.01
40734	Earth Fault Reclose Time 1 (S)	50-60000	2 Bytes	1	0.01
40735	Earth Fault Reclose Time 2 (S)	50-60000	2 Bytes	1	0.01
40736	Earth Fault Reclose Time 3 (S)	50-60000	2 Bytes	1	0.01
40737	Earth Fault Reclose Time 4 (S)	50-60000	2 Bytes	1	0.01
40738	Sensitive Earth Fault Reclose Time 1 (S)	50-60000	2 Bytes	1	0.01
40739	Sensitive Earth Fault Reclose Time 2 (S)	50-60000	2 Bytes	1	0.01
40740	Sensitive Earth Fault Reclose Time 3 (S)	50-60000	2 Bytes	1	0.01
40741	Sensitive Earth Fault Reclose Time 4 (S)	50-60000	2 Bytes	1	0.01
40742	AR 1 Protection Block	0-8191	2 Bytes	-	1
40743	AR 2 Protection Block	0-8191	2 Bytes	-	1
40744	AR 3 Protection Block	0-8191	2 Bytes	-	1
40745	AR 4 Protection Block	0-8191	2 Bytes	-	1
40746	Protection AR 1 Block	0-127	2 Bytes	-	1
40747	Protection AR 2 Block	0-127	2 Bytes	-	1
40748	Protection AR 3 Block	0-127	2 Bytes	-	1
40749	Protection AR 4 Block	0-127	2 Bytes	-	1

3.4 Cold Load Setting:

Address	Register Name	Range	Register Size	Step	Multiplicatio Factor
40750	Breaker OFF Time Hours	0-23	2 Bytes	1	1
40751	Breaker OFF Time Minutes	0-59	2 Bytes	1	1
40752	Breaker OFF Time Seconds	0-59	2 Bytes	1	1
40753	Breaker OFF Time Milliseconds	0-99	2 Bytes	1	1
40754	Cold Load Reset Time Hours	0-23	2 Bytes	1	1
40755	Cold Load Reset Time Minutes	0-59	2 Bytes	1	1
40756	Cold Load Reset Time Seconds	0-59	2 Bytes	1	1
40757	Cold Load Reset Time Milliseconds	0-99	2 Bytes	1	1
40758	Cold Load Reset (Time)	0-OFF , 1-ON	2 Bytes	1	1
40759	Cold Load Reset (Current)	0-OFF , 1-ON	2 Bytes	1	1

3.5 Demand Control Settings:

Address	Register Name	Range	Register Size	Step	Multiplicat Factor
40760	Max KVA Period	1-6	2 Bytes	1	1
40761	Max KVA Sub Period	1-60	2 Bytes	1	1
40762	Max KVA Setting	0-65000	2 Bytes	1	1
40763	Max KW Period	1-6	2 Bytes	1	1
40764	Max KW Sub Period	1-60	2 Bytes	1	1
40765	Max KW Setting	0-65000	2 Bytes	1	1

3.6 CB Maintaince Settings:

Address	Register Name	Range	Register Size	Step	Multiplicatio Factor
40766	Trip Count Alarm	0-999	2 Bytes	1	1
40767	Sum Of I^2 Alarm	0-65000	2 Bytes	1	1

3.7 Sectionalizer Settings:

Address	Register Name	Range	Register Size	Step	Multiplicatio Factor
40768	Fault Count	1-4	2 Bytes	1	1
40769	Reclaim Time (M)	0-6000	2 Bytes	1	0.01

40770	Auto Direction Group Set	0-OFF, 1-ON	2 Bytes	1	1
40771	AGBD Time (S)	1000-3000	2 Bytes	1	0.01

3.8 Output Relay Settings:

Address	Register Name	Range	Register Size	Step	Multi Fa
40772	Output Relay Configuration	0-OFF , 1-ON	2 Bytes	1	
40773	Hand Reset	0-31	2 Bytes	-	
40774	Non Trip Alarm	0-31	2 Bytes	-	
40775	Trip Alarm	0-31	2 Bytes	-	
40776	Phase Fault IDMT Start	0-31	2 Bytes	-	
40777	Phase Fault Wideset Start	0-31	2 Bytes	-	
40778	Earth Fault IDMT Start	0-31	2 Bytes	-	
40779	Earth Fault Wideset Start	0-31	2 Bytes	-	
40780	Phase Fault IDMT Trip	0-31	2 Bytes	-	
40781	Phase Fault Wideset Trip	0-31	2 Bytes	-	
40782	Phase Fault Highset Trip	0-31	2 Bytes	-	
40783	Earth Fault IDMT Trip	0-31	2 Bytes	-	
40784	Earth Fault Wideset Trip	0-31	2 Bytes	-	
40785	Earth Fault Highset Trip	0-31	2 Bytes	-	
40786	Over Voltage Stage1 Start	0-31	2 Bytes	-	
40787	Over Voltage Stage2 Start	0-31	2 Bytes	-	
40788	Under Voltage Stage1 Start	0-31	2 Bytes	-	
40789	Under Voltage Stage2 Start	0-31	2 Bytes	-	
40790	Residual Voltage Stage1 Start	0-31	2 Bytes	-	
40791	Residual Voltage Stage2 Start	0-31	2 Bytes	-	
40792	Over Voltage Stage1 Trip	0-31	2 Bytes	-	
40793	Over Voltage Stage2 Trip	0-31	2 Bytes	-	
40794	Under Voltage Stage1 Trip	0-31	2 Bytes	-	
40795	Under Voltage Stage2 Trip	0-31	2 Bytes	-	
40796	Residual Voltage Stage1 Trip	0-31	2 Bytes	-	
40797	Residual Voltage Stage2 Trip	0-31	2 Bytes	-	
40798	Sensitive Earth Fault Start	0-31	2 Bytes	-	
40799	Sensitive Earth Fault Trip	0-31	2 Bytes	-	
40800	Circuit Breaker Fail	0-31	2 Bytes	-	
40801	Trip Count Alarm	0-31	2 Bytes	-	
40802	Sum of I^2 Alarm	0-31	2 Bytes	-	
40803	Auto Reclose	0-31	2 Bytes	-	

40804	Reclaim Time	0-31	2 Bytes	-	
40805	Reclose Time	0-31	2 Bytes	-	
40806	Lock Out	0-31	2 Bytes	-	
40807	Trip Locked	0-31	2 Bytes	-	
40808	DC Healthy	0-31	2 Bytes	-	
40809	Spring Charge	0-31	2 Bytes	-	
40810	Trip Circuit Healthy	0-31	2 Bytes	-	
40811	Local Mode	0-31	2 Bytes	-	
40812	Remote Mode	0-31	2 Bytes	-	
40813	Maximum Demand KW	0-31	2 Bytes	-	
40814	Maximum Demand KVA	0-31	2 Bytes	-	
40815	Digital Input 5 Pickup	0-31	2 Bytes	-	
40816	Digital Input 6 Pickup	0-31	2 Bytes	-	
40817	Digital Input 7 Pickup	0-31	2 Bytes	-	
40818	Digital Input 8 Pickup	0-31	2 Bytes	-	
40819	Digital Input 9 Pickup	0-31	2 Bytes	-	
40820	Digital Input 10 Pickup	0-31	2 Bytes	-	
40821	Digital Input 11 Pickup	0-31	2 Bytes	-	
40822	Digital Input 12 Pickup	0-31	2 Bytes	-	
40823	Digital Input 13 Pickup	0-31	2 Bytes	-	
40824	Digital Input 14 Pickup	0-31	2 Bytes	-	
40825	Digital Input 15 Pickup	0-31	2 Bytes	-	
40826	Digital Input 5 DropOff	0-31	2 Bytes	-	
40827	Digital Input 6 DropOff	0-31	2 Bytes	-	
40828	Digital Input 7 DropOff	0-31	2 Bytes	-	
40829	Digital Input 8 DropOff	0-31	2 Bytes	-	
40830	Digital Input 9 DropOff	0-31	2 Bytes	-	
40831	Digital Input 10 DropOff	0-31	2 Bytes	-	
40832	Digital Input 11 DropOff	0-31	2 Bytes	-	
40833	Digital Input 12 DropOff	0-31	2 Bytes	-	
40834	Digital Input 13 DropOff	0-31	2 Bytes	-	
40835	Digital Input 14 DropOff	0-31	2 Bytes	-	
40836	Digital Input 15 DropOff	0-31	2 Bytes	-	
40837	Push Button 1 Pickup	0-31	2 Bytes	-	
40838	Push Button 2 Pickup	0-31	2 Bytes	-	
40839	Push Button 1 DropOff	0-31	2 Bytes	-	
40840	Push Button 2 DropOff	0-31	2 Bytes	-	
40841	R Phase Trip	0-31	2 Bytes	-	
40842	Y Phase Trip	0-31	2 Bytes	-	
40843	B Phase Trip	0-31	2 Bytes	-	
40844	Realtime RLY1	0-31	2 Bytes	-	
40845	Realtime RLY2	0-31	2 Bytes	-	

3.9 LED Settings:

Address	Register Name	Range	Register Size	Step	Multiplic Factor
40846	LED Configuration	0-OFF, 1-ON	2 Bytes	1	1
40847	R Phase IDMT/DTL Start	*Note 3	2 Bytes	-	1
40848	Y Phase IDMT/DTL Start	*Note 3	2 Bytes	-	1
40849	B Phase IDMT/DTL Start	*Note 3	2 Bytes	-	1
40850	R Wideset Start	*Note 3	2 Bytes	-	1
40851	Y Wideset Start	*Note 3	2 Bytes	-	1
40852	B Wideset Start	*Note 3	2 Bytes	-	1
40853	Earth IDMT/DTL Start	*Note 3	2 Bytes	-	1
40854	Earth Wideset Start	*Note 3	2 Bytes	-	1
40855	R Phase IDMT/DTL Trip	*Note 3	2 Bytes	-	1
40856	Y Phase IDMT/DTL Trip	*Note 3	2 Bytes	-	1
40857	B Phase IDMT/DTL Trip	*Note 3	2 Bytes	-	1
40858	R Phase Wideset Trip	*Note 3	2 Bytes	-	1
40859	Y Phase Wideset Trip	*Note 3	2 Bytes	-	1
40860	B Phase Wideset Trip	*Note 3	2 Bytes	-	1
40861	R Phase Highset Trip	*Note 3	2 Bytes	-	1
40862	Y Phase Highset Trip	*Note 3	2 Bytes	-	1
40863	B Phase Highset Trip	*Note 3	2 Bytes	-	1
40864	Earth IDMT/DTL Trip	*Note 3	2 Bytes	-	1
40865	Earth Wideset Trip	*Note 3	2 Bytes	-	1
40866	Earth Highset Trip	*Note 3	2 Bytes	-	1
40867	RN OV Stage1 Start	*Note 3	2 Bytes	-	1
40868	YN OV Stage1 Start	*Note 3	2 Bytes	-	1
40869	BN OV Stage1 Start	*Note 3	2 Bytes	-	1
40870	RN OV Stage2 Start	*Note 3	2 Bytes	-	1
40871	YN OV Stage2 Start	*Note 3	2 Bytes	-	1
40872	BN OV Stage2 Start	*Note 3	2 Bytes	-	1
40873	RN UV Stage1 Start	*Note 3	2 Bytes	-	1
40874	YN UV Stage1 Start	*Note 3	2 Bytes	-	1
40875	BN UV Stage1 Start	*Note 3	2 Bytes	-	1
40876	RN UV Stage2 Start	*Note 3	2 Bytes	-	1
40877	YN UV Stage2 Start	*Note 3	2 Bytes	-	1
40878	BN UV Stage2 Start	*Note 3	2 Bytes	-	1
40879	RN OV Stage 1 Trip	*Note 3	2 Bytes	-	1
40880	YN OV Stage 1 Trip	*Note 3	2 Bytes	-	1
40881	BN OV Stage 1 Trip	*Note 3	2 Bytes	-	1
40882	RN OV Stage 2 Trip	*Note 3	2 Bytes	-	1
40883	YN OV Stage 2 Trip	*Note 3	2 Bytes	-	1
40884	BN OV Stage 2 Trip	*Note 3	2 Bytes	-	1
40885	RN UV Stage 1 Trip	*Note 3	2 Bytes	-	1
40886	YN UV Stage 1 Trip	*Note 3	2 Bytes	-	1
40887	BN UV Stage 1 Trip	*Note 3	2 Bytes	-	1

40888	RN UV Stage 2 Trip	*Note 3	2 Bytes	-	1
40889	YN UV Stage 2 Trip	*Note 3	2 Bytes	-	1
40890	BN UV Stage 2 Trip	*Note 3	2 Bytes	-	1
40891	Residual Voltage Stage 1 Start	*Note 3	2 Bytes	-	1
40892	Residual Voltage Stage 2 Start	*Note 3	2 Bytes	-	1
40893	Residual Voltage Stage 1 Trip	*Note 3	2 Bytes	-	1
40894	Residual Voltage Stage 2 Trip	*Note 3	2 Bytes	-	1
40895	Sensitive Earth Fault Start	*Note 3	2 Bytes	-	1
40896	Sensitive Earth Fault Trip	*Note 3	2 Bytes	-	1
40897	Circuit Breaker Failure	*Note 3	2 Bytes	-	1
40898	Phase Fault Reclose Time1	*Note 3	2 Bytes	-	1
40899	Phase Fault Reclose Time2	*Note 3	2 Bytes	-	1
40900	Phase Fault Reclose Time3	*Note 3	2 Bytes	-	1
40901	Phase Fault Reclose Time4	*Note 3	2 Bytes	-	1
40902	Earth Fault Reclose Time1	*Note 3	2 Bytes	-	1
40903	Earth Fault Reclose Time2	*Note 3	2 Bytes	-	1
40904	Earth Fault Reclose Time3	*Note 3	2 Bytes	-	1
40905	Earth Fault Reclose Time4	*Note 3	2 Bytes	-	1
40906	Sensitive Earth Fault Reclose Time 1	*Note 3	2 Bytes	-	1
40907	Sensitive Earth Fault Reclose Time 2	*Note 3	2 Bytes	-	1
40908	Sensitive Earth Fault Reclose	*Note 3	2 Bytes	-	1
	Time 3				
40909	Sensitive Earth Fault Reclose Time 4	*Note 3	2 Bytes	-	1
40910	Reclaim Time	*Note 3	2 Bytes	-	1
40911	AR 1 Protection Block	*Note 3	2 Bytes	-	1
40912	AR 2 Protection Block	*Note 3	2 Bytes	-	1
40913	AR 3 Protection Block	*Note 3	2 Bytes	-	1
40914	AR 4 Protection Block	*Note 3	2 Bytes	-	1
40915	Protection AR 1 Block	*Note 3	2 Bytes	-	1
40916	Protection AR 2 Block	*Note 3	2 Bytes	-	1
40917	Protection AR 3 Block	*Note 3	2 Bytes	-	1
40918	Protection AR 4 Block	*Note 3	2 Bytes	-	1
40919	Lock Out	*Note 3	2 Bytes	-	1
40920	Trip Count Alarm	*Note 3	2 Bytes	-	1
40921	Sum of I^2 Alarm	*Note 3	2 Bytes	-	1
40922	Maximum Demand KW	*Note 3	2 Bytes	-	1
40923	Maximum Demand KVA	*Note 3	2 Bytes	-	1
40924	Digital Input 5 Pickup	*Note 3	2 Bytes	-	1
40925	Digital Input 6 Pickup	*Note 3	2 Bytes	-	1
40926	Digital Input 7 Pickup	*Note 3	2 Bytes	-	1
40927	Digital Input 8 Pickup	*Note 3	2 Bytes	-	1
40928	Digital Input 9 Pickup	*Note 3	2 Bytes	-	1

40929	Digital Input 10 Pickup	*Note 3	2 Bytes	-	1
40930	Digital Input 11 Pickup	*Note 3	2 Bytes	-	1
40931	Digital Input 12 Pickup	*Note 3	2 Bytes	-	1
40932	Digital Input 13 Pickup	*Note 3	2 Bytes	-	1
40933	Digital Input 14 Pickup	*Note 3	2 Bytes	-	1
40934	Digital Input 15 Pickup	*Note 3	2 Bytes	-	1
40935	Digital Input 5 DropOff	*Note 3	2 Bytes	-	1
40936	Digital Input 6 DropOff	*Note 3	2 Bytes	-	1
40937	Digital Input 7 DropOff	*Note 3	2 Bytes	-	1
40938	Digital Input 8 DropOff	*Note 3	2 Bytes	-	1
40939	Digital Input 9 DropOff	*Note 3	2 Bytes	-	1
40940	Digital Input 10 DropOff	*Note 3	2 Bytes	-	1
40941	Digital Input 11 DropOff	*Note 3	2 Bytes	-	1
40942	Digital Input 12 DropOff	*Note 3	2 Bytes	-	1
40943	Digital Input 13 DropOff	*Note 3	2 Bytes	-	1
40944	Digital Input 14 DropOff	*Note 3	2 Bytes	-	1
40945	Digital Input 15 DropOff	*Note 3	2 Bytes	-	1
40946	Active Group 1	*Note 3	2 Bytes	-	1
40947	Active Group 2	*Note 3	2 Bytes	-	1
40948	Active Group 3	*Note 3	2 Bytes	-	1
40949	Active Group 4	*Note 3	2 Bytes	-	1
40950	Active Group 5	*Note 3	2 Bytes	-	1
40951	Active Group 6	*Note 3	2 Bytes	-	1
40952	Active Group 7	*Note 3	2 Bytes	-	1
40953	Active Group 8	*Note 3	2 Bytes	-	1
40954	Active Group 9	*Note 3	2 Bytes	-	1
40955	Active Group 10	*Note 3	2 Bytes	-	1
40956	Push Button 1 Pickup	*Note 3	2 Bytes	-	1
40957	Push Button 2 Pickup	*Note 3	2 Bytes	-	1
40958	Push Button 1 DropOff	*Note 3	2 Bytes	-	1
40959	Push Button 2 DropOff	*Note 3	2 Bytes	-	1
40960	Realtime RLY1	*Note 3	2 Bytes	-	1
40961	Realtime RLY2	*Note 3	2 Bytes	-	1
40962	Red LED Reset	*Note 3.1	2 Bytes	-	1
	Yellow LED Reset				

Note 3:

- Yellow LED (Low Byte) – Only one LED can be set for a Setting
- Red LED (High Byte) – Only one LED can be set for a Setting

We can set either one Yellow LED or one Red LED for a setting we cannot set both Yellow LED and Red LED

Note 3.1:

- Yellow LED (Low Byte) – More than one LED can be set for this Setting
- Red LED (High Byte) – More than one LED can be set for the Setting

In this setting we can set both the Yellow and Red LED at the same time the values ranges from 0-255 for each Yellow LEDs and eight Red LEDs at the same time

3.10 Digital Input Timer:

Address	Register Name	Range	Register Size	Step	Multiplic Facto
40963	Digital Input1 Pickup Hours Digital Input1 Pickup Minutes	0-23 0-59	2 Bytes	1	1
40964	Digital Input1 Pickup Seconds Digital Input1 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
40965	Digital Input1 DropOff Hours Digital Input1 DropOff Minutes	0-23 0-59	2 Bytes	1	1
40966	Digital Input1 DropOff Seconds Digital Input1 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
40967	Digital Input2 Pickup Hours Digital Input2 Pickup Minutes	0-23 0-59	2 Bytes	1	1
40968	Digital Input2 Pickup Seconds Digital Input2 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
40969	Digital Input2 DropOff Hours Digital Input2 DropOff Minutes	0-23 0-59	2 Bytes	1	1
40970	Digital Input2 DropOff Seconds Digital Input2 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
40971	Digital Input3 Pickup Hours Digital Input3 Pickup Minutes	0-23 0-59	2 Bytes	1	1
40972	Digital Input3 Pickup Seconds Digital Input3 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40973	Digital Input3 DropOff Hours Digital Input3 DropOff Minutes	0-23 0-59	2 Bytes	1	
40974	Digital Input3 DropOff Seconds Digital Input3 DropOff Milliseconds	0-59 0-99	2 Bytes	1	
40975	Digital Input4 Pickup Hours Digital Input4 Pickup Minutes	0-23 0-59	2 Bytes	1	
40976	Digital Input4 Pickup Seconds Digital Input4 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40977	Digital Input4 DropOff Hours Digital Input4 DropOff Minutes	0-23 0-59	2 Bytes	1	
40978	Digital Input4 DropOff Seconds Digital Input4 DropOff Milliseconds	0-59 0-99	2 Bytes	1	

40979	Digital Input5 Pickup Hours Digital Input5 Pickup Minutes	0-23 0-59	2 Bytes	1	
40980	Digital Input5 Pickup Seconds Digital Input5 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40981	Digital Input5 DropOff Hours Digital Input5 DropOff Minutes	0-23 0-59	2 Bytes	1	
40982	Digital Input5 DropOff Seconds Digital Input5 DropOff Milliseconds	0-59 0-99	2 Bytes	1	
40983	Digital Input6 Pickup Hours Digital Input6 Pickup Minutes	0-23 0-59	2 Bytes	1	
40984	Digital Input6 Pickup Seconds Digital Input6 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40985	Digital Input6 DropOff Hours Digital Input6 DropOff Minutes	0-23 0-59	2 Bytes	1	
40986	Digital Input6 DropOff Seconds Digital Input6 DropOff Milliseconds	0-59 0-99	2 Bytes	1	
40987	Digital Input7 Pickup Hours Digital Input7 Pickup Minutes	0-23 0-59	2 Bytes	1	
40988	Digital Input7 Pickup Seconds Digital Input7 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40989	Digital Input7 DropOff Hours Digital Input7 DropOff Minutes	0-23 0-59	2 Bytes	1	
40990	Digital Input7 DropOff Seconds Digital Input7 DropOff Milliseconds	0-59 0-99	2 Bytes	1	
40991	Digital Input8 Pickup Hours Digital Input8 Pickup Minutes	0-23 0-59	2 Bytes	1	
40992	Digital Input8 Pickup Seconds Digital Input8 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
40993	Digital Input8 DropOff Hours Digital Input8 DropOff Minutes	0-23 0-59	2 Bytes	1	
40994	Digital Input8 DropOff Seconds Digital Input8 DropOff Milliseconds	0-59 0-99	2 Bytes	1	
40995	Digital Input9 Pickup Hours Digital Input9 Pickup Minutes	0-23 0-59	2 Bytes	1	
40996	Digital Input9 Pickup Seconds Digital Input9 Pickup Milliseconds	0-59 0-99	2 Bytes	1	
40997	Digital Input9 DropOff Hours Digital Input9 DropOff Minutes	0-23 0-59	2 Bytes	1	1
40998	Digital Input9 DropOff Seconds Digital Input9 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
40999	Digital Input10 Pickup Hours Digital Input10 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41000	Digital Input10 Pickup Seconds Digital Input10 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1

41001	Digital Input10 DropOff Hours Digital Input10 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41002	Digital Input10 DropOff Seconds Digital Input10 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41003	Digital Input11 Pickup Hours Digital Input11 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41004	Digital Input11 Pickup Seconds Digital Input11 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41005	Digital Input11 DropOff Hours Digital Input11 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41006	Digital Input11 DropOff Seconds Digital Input11 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41007	Digital Input12 Pickup Hours Digital Input12 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41008	Digital Input12 Pickup Seconds Digital Input12 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41009	Digital Input12 DropOff Hours Digital Input12 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41010	Digital Input12 DropOff Seconds Digital Input12 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41011	Digital Input13 Pickup Hours Digital Input13 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41012	Digital Input13 Pickup Seconds Digital Input13 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41013	Digital Input13 DropOff Hours Digital Input13 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41014	Digital Input13 DropOff Seconds Digital Input13 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41015	Digital Input14 Pickup Hours Digital Input14 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41016	Digital Input14 Pickup Seconds Digital Input14 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41017	Digital Input14 DropOff Hours Digital Input14 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41018	Digital Input14 DropOff Seconds Digital Input14 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41019	Digital Input15 Pickup Hours Digital Input15 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41020	Digital Input15 Pickup Seconds Digital Input15 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41021	Digital Input15 DropOff Hours Digital Input15 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41022	Digital Input15 DropOff Seconds Digital Input15 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1

41023	Push Button1 Pickup Hours Push Button1 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41024	Push Button1 Pickup Seconds Push Button1 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41025	Push Button1 DropOff Hours Push Button1 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41026	Push Button1 DropOff Seconds Push Button1 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1
41027	Push Button2 Pickup Hours Push Button2 Pickup Minutes	0-23 0-59	2 Bytes	1	1
41028	Push Button2 Pickup Seconds Push Button2 Pickup Milliseconds	0-59 0-99	2 Bytes	1	1
41029	Push Button2 DropOff Hours Push Button2 DropOff Minutes	0-23 0-59	2 Bytes	1	1
41030	Push Button2 DropOff Seconds Push Button2 DropOff Milliseconds	0-59 0-99	2 Bytes	1	1

3.11 Digital Input Blocking:

Address	Register Name	Range (HEX)	Register Size	Step	Multiplicati Factor
41031	Digital Input1 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41032	Digital Input1 Pickup DI Block	0-7FFE	2 Bytes	-	-
41033	Digital Input1 Pickup Control Block	0-001F	2 Bytes	-	-
41034	Digital Input1 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41035	Digital Input1 DropOff DI Block	0-7FFE	2 Bytes	-	-
41036	Digital Input1 DropOff Control Block	0-001F	2 Bytes	-	-
41037	Digital Input2 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41038	Digital Input2 Pickup DI Block	0-7FFD	2 Bytes	-	-
41039	Digital Input2 Pickup Control Block	0-001F	2 Bytes	-	-
41040	Digital Input2 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41041	Digital Input2 DropOff DI Block	0-7FFD	2 Bytes	-	-
41042	Digital Input2 DropOff Control Block	0-001F	2 Bytes	-	-
41043	Digital Input3 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41044	Digital Input3 Pickup DI Block	0-7FFB	2 Bytes	-	-
41045	Digital Input3 Pickup Control Block	0-001F	2 Bytes	-	-
41046	Digital Input3 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41047	Digital Input3 DropOff DI Block	0-7FFB	2 Bytes	-	-
41048	Digital Input3 DropOff Control Block	0-001F	2 Bytes	-	-
41049	Digital Input4 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41050	Digital Input4 Pickup DI Block	0-7FF7	2 Bytes	-	-
41051	Digital Input4 Pickup Control Block	0-001F	2 Bytes	-	-

41052	Digital Input4 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41053	Digital Input4 DropOff DI Block	0-7FF7	2 Bytes	-	-
41054	Digital Input4 DropOff Control Block	0-001F	2 Bytes	-	-
41055	Digital Input5 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41056	Digital Input5 Pickup DI Block	0-7FEF	2 Bytes	-	-
41057	Digital Input5 Pickup Control Block	0-001F	2 Bytes	-	-
41058	Digital Input5 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41059	Digital Input5 DropOff DI Block	0-7FEF	2 Bytes	-	-
41060	Digital Input5 DropOff Control Block	0-001F	2 Bytes	-	-
41061	Digital Input6 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41062	Digital Input6 Pickup DI Block	0-7FDF	2 Bytes	-	-
41063	Digital Input6 Pickup Control Block	0-001F	2 Bytes	-	-
41064	Digital Input6 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41065	Digital Input6 DropOff DI Block	0-7FDF	2 Bytes	-	-
41066	Digital Input6 DropOff Control Block	0-001F	2 Bytes	-	-
41067	Digital Input7 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41068	Digital Input7 Pickup DI Block	0-7FBF	2 Bytes	-	-
41069	Digital Input7 Pickup Control Block	0-001F	2 Bytes	-	-
41070	Digital Input7 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41071	Digital Input7 DropOff DI Block	0-7FBF	2 Bytes	-	-
41072	Digital Input7 DropOff Control Block	0-001F	2 Bytes	-	-
41073	Digital Input8 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41074	Digital Input8 Pickup DI Block	0-7F7F	2 Bytes	-	-
41075	Digital Input8 Pickup Control Block	0-001F	2 Bytes	-	-
41076	Digital Input8 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41077	Digital Input8 DropOff DI Block	0-7F7F	2 Bytes	-	-
41078	Digital Input8 DropOff Control Block	0-001F	2 Bytes	-	-
41079	Digital Input9 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41080	Digital Input9 Pickup DI Block	0-7EFF	2 Bytes	-	-
41081	Digital Input9 Pickup Control Block	0-001F	2 Bytes	-	-
41082	Digital Input9 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41083	Digital Input9 DropOff DI Block	0-7EFF	2 Bytes	-	-
41084	Digital Input9 DropOff Control Block	0-001F	2 Bytes	-	-
41085	Digital Input10 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41086	Digital Input10 Pickup DI Block	0-7DFF	2 Bytes	-	-
41087	Digital Input10 Pickup Control Block	0-001F	2 Bytes	-	-
41088	Digital Input10 DropOff Protection Block	0-1FFF	2 Bytes	-	-

41089	Digital Input10 DropOff DI Block	0-7DFF	2 Bytes	-	-
41090	Digital Input10 DropOff Control Block	0-001F	2 Bytes	-	-
41091	Digital Input11 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41092	Digital Input11 Pickup DI Block	0-7BFF	2 Bytes	-	-
41093	Digital Input11 Pickup Control Block	0-001F	2 Bytes	-	-
41094	Digital Input11 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41095	Digital Input11 DropOff DI Block	0-7BFF	2 Bytes	-	-
41096	Digital Input11 DropOff Control Block	0-001F	2 Bytes	-	-
41097	Digital Input12 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41098	Digital Input12 Pickup DI Block	0-77FF	2 Bytes	-	-
41099	Digital Input12 Pickup Control Block	0-001F	2 Bytes	-	-
41100	Digital Input12 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41101	Digital Input12 DropOff DI Block	0-77FF	2 Bytes	-	-
41102	Digital Input12 DropOff Control Block	0-001F	2 Bytes	-	-
41103	Digital Input13 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41104	Digital Input13 Pickup DI Block	0-6FFF	2 Bytes	-	-
41105	Digital Input13 Pickup Control Block	0-001F	2 Bytes	-	-
41106	Digital Input13 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41107	Digital Input13 DropOff DI Block	0-6FFF	2 Bytes	-	-
41108	Digital Input13 DropOff Control Block	0-001F	2 Bytes	-	-
41109	Digital Input14 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41110	Digital Input14 Pickup DI Block	0-5FFF	2 Bytes	-	-
41111	Digital Input14 Pickup Control Block	0-001F	2 Bytes	-	-
41112	Digital Input14 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41113	Digital Input14 DropOff DI Block	0-5FFF	2 Bytes	-	-
41114	Digital Input14 DropOff Control Block	0-001F	2 Bytes	-	-
41115	Digital Input15 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41116	Digital Input15 Pickup DI Block	0-3FFF	2 Bytes	-	-
41117	Digital Input15 Pickup Control Block	0-001F	2 Bytes	-	-
41118	Digital Input15 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41119	Digital Input15 DropOff DI Block	0-3FFF	2 Bytes	-	-
41120	Digital Input15 DropOff Control Block	0-001F	2 Bytes	-	-
41121	Push Button1 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41122	Push Button1 Pickup DI Block	0-7FFF	2 Bytes	-	-
41123	Push Button1 Pickup Control Block	0-001E	2 Bytes	-	-

41124	Push Button1 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41125	Push Button1 DropOff DI Block	0-7FFF	2 Bytes	-	-
41126	Push Button1 DropOff Control Block	0-001D	2 Bytes	-	-
41127	Push Button2 Pickup Protection Block	0-1FFF	2 Bytes	-	-
41128	Push Button2 Pickup DI Block	0-7FFF	2 Bytes	-	-
41129	Push Button2 Pickup Control Block	0-001E	2 Bytes	-	-
41130	Push Button2 DropOff Protection Block	0-1FFF	2 Bytes	-	-
41131	Push Button2 DropOff DI Block	0-7FFF	2 Bytes	-	-
41132	Push Button2 DropOff Control Block	0-001D	2 Bytes	-	-
41133	Digital Input Invert	0-07FF	2 Bytes	-	-

3.12 LED STATUS:

Address	Register Name	Value	Register Size	Step	Multiplication Factor
41134	LED Status 1	0-255	2 Bytes	-	1
41135	LED Status 2	0-255	2 Bytes	-	1
41136	LED Status 3	0-255	2 Bytes	-	1
41137	Flash LED 1	0-255	2 Bytes	-	1
41138	Flash LED 2	0-255	2 Bytes	-	1
41139	Accept Value	0-255	2 Bytes	-	1
41140	Reset Value	0-255	2 Bytes	-	1
41141	Relay Status	0-1F	2 Bytes	-	1

4 COMMANDS:

Address	Register Name	Value	Register Size	Step	Multiplication Factor
40219	Remote Trip	1	2 Bytes	-	1
	Remote Close	2		-	1
40220	Push Button 1 ON	2	2 Bytes	-	1
	Push Button 1 OFF	1		-	1
40221	Push Button 2 ON	2	2 Bytes	-	1
	Push Button 2 OFF	1		-	1
40222	Device Reset Remote	2	2 Bytes	-	1
40223	Fault Record Clear Remote	2	2 Bytes	-	1
40224	Event Record Clear Remote	2	2 Bytes	-	1
40225	Trip Count Clear Remote	2	2 Bytes	-	1
40226	I^2 Alarm Clear Remote	2	2 Bytes	-	1
40227	KVA Clear	2	2 Bytes	-	1
40228	KW Clear	2	2 Bytes	-	1
40229	Outage Bus Clear	2	2 Bytes	-	1
40230	Outage Load Clear	2	2 Bytes	-	1

5 TIME SYNCHRONIZATION:

Address	Register Name	Range	Register Size	Step	Multiplication Factor
40281	Present Seconds	0-59	1 Byte	-	1
	Present Minutes	0-59	1 Byte	-	1
40282	Present Hours	0-23	1 Byte	-	1
	Present Day	1-31	1 Byte	-	1
40283	Present Month	1-12	1 Byte	-	1
	Present Year	0-99	1 Byte	-	1

6 FAULT RECORD:

Address	Register Name	Range	Register Size	Step	Multiplicatio Factor
41201-41202	R Phase Fault Current	-	4 Bytes	-	0.01
41203-41204	Y Phase Fault Current	-	4 Bytes	-	0.01
41205-41206	B Phase Fault Current	-	4 Bytes	-	0.01
41207-41208	Earth Fault Current	-	4 Bytes	-	0.01
41209-41210	Sensitive Earth Fault Current	-	4 Bytes	-	0.01
41211-41212	I1 Positive Sequence Current	-	4 Bytes	-	0.01
41213-41214	I2 Negative Sequence Current	-	4 Bytes	-	0.01
41215-41216	I0 Zero Sequence Current	-	4 Bytes	-	0.01
41217-41218	R Phase Fault Voltage	-	4 Bytes	-	0.0001
41219-41220	Y Phase Fault Voltage	-	4 Bytes	-	0.0001
41221-41222	B Phase Fault Voltage	-	4 Bytes	-	0.0001
41223-41224	Residual Fault Voltage	-	4 Bytes	-	0.0001
41225-41226	V1 Positive Sequence Voltage	-	4 Bytes	-	0.0001
41227-41228	V2 Negative Sequence Voltage	-	4 Bytes	-	0.0001
41229-41230	V0 Zero Sequence Voltage	-	4 Bytes	-	0.0001
41231	Current Start Element	A	2 Bytes	-	1
41232	Current Trip Element	B	2 Bytes	-	1
41233	Future	-	2 Bytes	-	1
41234	Voltage Start Element	C	2 Bytes	-	1
41235	Voltage Trip Element	D	2 Bytes	-	1
41236	Residual Voltage Trip	E	2 Bytes	-	1
41237	Auto Reclose Shots	1-4	1 Byte	-	1
	Trip Group	1-10	1 Byte	-	1
41238	Trip Time Milliseconds	0-999	2 Bytes	-	1
41239	Trip Time Minutes	0-59	1 Byte	-	1
	Trip Time Hours	0.23	1 Byte	-	1
41240	Trip Time Day	1-31	1 Byte	-	1
	Trip Date Seconds	0-59	1 Byte	-	1
41241	Trip Date Year	0-99	1 Byte	-	1
	Trip Date Month	1-12	1 Byte	-	1
41242	Fault Record Pointer	-	2 Bytes	-	1

41243-41244	Reclaim Time	-	4 Bytes	-	1

Trip Element Message:

Current Start Element (Reg-41231):

	1	1- Phase Curve Type
	2	2- R_Ph_IDMT_Starter
	3	3- Y_Ph_IDMT_Starter
	4	4- B_Ph_IDMT_Starter
	5	5- Phase Wideset Starter
	6	6- R_Ph_Wideset Starter
	7	7- Y_Ph_Wideset Starter
A	8	8- B_Ph_Wideset Starter
	9	9- Phase IDMT Starter
	10	10- Earth Curve Type
	11	11- Earth IDMT Starter
	12	12- Earth Wideset Starter
	13	13- Phase Reset Starter
	14	14- Earth Reset Starter
	15	15- Auto Lockout
	16	16- SEF Starter

Current Trip Element (Reg-41232):

B	1	1- SEF Trip
	2	2- R_Ph_IDMT_Trip
	3	3- Y_Ph_IDMT_Trip
	4	4- B_Ph_IDMT_Trip
	5	5- Phase IDMT Trip
	6	6- R_Ph_Wideset Trip
	7	7- Y_Ph_Wideset Trip
	8	8- B_Ph_Wideset Trip
	9	9- Phase Wideset Trip
	10	10- R_Highset Trip
	11	11- Y_Highset Trip
	12	12- B_Highset Trip
	13	13- Trip Enable
	14	14- Earth IDMT Trip
	15	15- Earth Wideset Trip
	16	16- Earth Highset Trip

Voltage Start Element (Reg-41234):

	1	1- Inv OV1 Type

C	2	2- Inv UV1 Type
	3	3- RN_Inv_OV1_Starter
	4	4- YN_Inv_OV1_Starter
	5	5- BN_Inv_OV1_Starter
	6	6- RN_Inv_UV1_Starter
	7	7- YN_Inv_UV1_Starter
	8	8- BN_Inv_UV1_Starter
	9	9- RN_DTL_OV2_Starter
	10	10- YN_DTL_OV2_Starter
	11	11- BN_DTL_OV2_Starter
	12	12- RN_DTL_UV2_Starter
	13	13- YN_DTL_UV2_Starter
	14	14- BN_DTL_UV2_Starter
	15	15- NE_Inv_RV1_Starter
	16	16- Inv_RV1_Type

Voltage Trip Element (Reg-41235):

D	1	1- NE_DTL_RV2_Starter
	2	2- OV_Rst_Starter
	3	3- UV_Rst_Starter
	4	4- RV_Rst_Starter
	5	5- RN_Inv_OV1_Trip
	6	6- YN_Inv_OV1_Trip
	7	7- BN_Inv_OV1_Trip
	8	8- RN_Inv_UV1_Trip
	9	9- YN_Inv_UV1_Trip
	10	10- BN_Inv_UV1_Trip
	11	11- RN_DTL_OV2_Trip
	12	12- YN_DTL_OV2_Trip
	13	13- BN_DTL_OV2_Trip
	14	14- RN_DTL_UV2_Trip
	15	15- YN_DTL_UV2_Trip
	16	16- BN_DTL_UV2_Trip

E	1	1- NE_Inv_RV1_Trip
	2	2- NE_DTL_RV2_Trip

7 EVENT RECORD:

Address	Register Name	Range	Register Size	Step	Multiplication Factor
45602	Event Record Pointer	-	2 Bytes	-	1
45603	Event Register A	-	2 Bytes	-	1
45604	Event Register B	-	2 Bytes	-	1
45605	Event Register C	-	2 Bytes	-	1

45606	Event Register D	-	2 Bytes	-	1
45607	Event Register E	-	2 Bytes	-	1
45608	Event Register F	-	2 Bytes	-	1
45609	Event Register G	-	2 Bytes	-	1
45610	Event Register H	-	2 Bytes	-	1
45611	Event Register I	-	2 Bytes	-	1
45612	Event Register J	-	2 Bytes	-	1
45613	Event Register K	-	2 Bytes	-	1
45614	Event Register L	-	2 Bytes	-	1
45615	Event Register M	-	2 Bytes	-	1
45616	DropOff Event Register A	-	2 Bytes	-	1
45617	DropOff Event Register B	-	2 Bytes	-	1
45618	DropOff Event Register C	-	2 Bytes	-	1
45619	DropOff Event Register D	-	2 Bytes	-	1
45620	DropOff Event Register E	-	2 Bytes	-	1
45621	DropOff Event Register F	-	2 Bytes	-	1
45622	DropOff Event Register G	-	2 Bytes	-	1
45623	DropOff Event Register H	-	2 Bytes	-	1
45624	DropOff Event Register I	-	2 Bytes	-	1
45625	DropOff Event Register J	-	2 Bytes	-	1
45626	DropOff Event Register K	-	2 Bytes	-	1
45267	DropOff Event Register L	-	2 Bytes	-	1
45628	DropOff Event Register M	-	2 Bytes	-	1
45629	Event Time Milliseconds	0-999	2 Bytes	-	1
45630	Event Time Hours	0-23	1 Bytes	-	1
	Event Time Minutes	0-59	1 Bytes	-	1
45631	Event Date Seconds	0-59	1 Bytes	-	1
	Event Time Day	1-31	1 Byte	-	1
45632	Event Date Year	0-99	1 Byte	-	1
	Event Date Month	1-12	1 Byte	-	1
45633-45634	Future	-	4 Bytes	-	-

8 EVENTS:

Address	Register Name	Range	Register Size	Step	Multiplication Factor
25001	Event Register A	-	2 Bytes	-	1
25002	Event Register B	-	2 Bytes	-	1
25003	Event Register C	-	2 Bytes	-	1
25004	Event Register D	-	2 Bytes	-	1
25005	Event Register E	-	2 Bytes	-	1
25006	Event Register F	-	2 Bytes	-	1
25007	Event Register G	-	2 Bytes	-	1
25008	Event Register H	-	2 Bytes	-	1
25009	Event Register I	-	2 Bytes	-	1

25010	Event Register J	-	2 Bytes	-	1
25011	Event Register K	-	2 Bytes	-	1
25012	Event Register L	-	2 Bytes	-	1
25013	Event Register M	-	2 Bytes	-	1

Event Register A:

	1	1- Phase Curve Type
	2	2- R_Ph_IDMT_Starter
	3	3- Y_Ph_IDMT_Starter
	4	4- B_Ph_IDMT_Starter
	5	5- Phase Wideset Starter
	6	6- R_Ph_Wideset Starter
	7	7- Y_Ph_Wideset Starter
	8	8- B_Ph_Wideset Starter
	9	9- Phase IDMT Starter
A	10	10- Earth Curve Type
	11	11- Earth IDMT Starter
	12	12- Earth Wideset Starter
	13	13- Phase Reset Starter
	14	14- Earth Reset Starter
	15	15- Auto Lockout
	16	16- SEF Starter

Event Register B:

	1	1- SEF Trip
	2	2- R_Ph_IDMT_Trip
	3	3- Y_Ph_IDMT_Trip
	4	4- B_Ph_IDMT_Trip
	5	5- Phase IDMT Trip
	6	6- R_Ph_Wideset Trip
	7	7- Y_Ph_Wideset Trip
	8	8- B_Ph_Wideset Trip
B	9	9- Phase Wideset Trip
	10	10- R_Highset Trip
	11	11- Y_Highset Trip
	12	12- B_Highset Trip
	13	13- Trip Enable
	14	14- Earth IDMT Trip
	15	15- Earth Wideset Trip
	16	16- Earth Highset Trip

Event Register C:

C	1	1- Inv OV1 Type
	2	2- Inv UV1 Type
	3	3- RN_Inv_OV1_Starter
	4	4- YN_Inv_OV1_Starter
	5	5- BN_Inv_OV1_Starter
	6	6- RN_Inv_UV1_Starter
	7	7- YN_Inv_UV1_Starter
	8	8- BN_Inv_UV1_Starter
	9	9- RN_DTL_OV2_Starter
	10	10- YN_DTL_OV2_Starter
	11	11- BN_DTL_OV2_Starter
	12	12- RN_DTL_UV2_Starter
	13	13- YN_DTL_UV2_Starter
	14	14- BN_DTL_UV2_Starter
	15	15- NE_Inv_RV1_Starter
	16	16- Inv_RV1_Type

Event Register D:

D	1	1- NE_DTL_RV2_Starter
	2	2- OV_Rst_Starter
	3	3- UV_Rst_Starter
	4	4- RV_Rst_Starter
	5	5- RN_Inv_OV1_Trip
	6	6- YN_Inv_OV1_Trip
	7	7- BN_Inv_OV1_Trip
	8	8- RN_Inv_UV1_Trip
	9	9- YN_Inv_UV1_Trip
	10	10- BN_Inv_UV1_Trip
	11	11- RN_DTL_OV2_Trip
	12	12- YN_DTL_OV2_Trip
	13	13- BN_DTL_OV2_Trip
	14	14- RN_DTL_UV2_Trip
	15	15- YN_DTL_UV2_Trip
	16	16- BN_DTL_UV2_Trip

Event Register E:

E	1	1- NE_Inv_RV1_Trip
	2	2- NE_DTL_RV2_Trip
	3	3- OV1 Trip
	4	4- UV1 Trip
	5	5- Power Flow Reverse
	6	6- Power Flow Forward
	7	7- RN_Input_Volt_ST
	8	8- RN_Input_Volt_ST20

E	9	9- RN_Input_Volt_ST80
	10	10- YN_Input_Volt_ST
	11	11- YN_Input_Volt_ST20
	12	12- YN_Input_Volt_ST80
	13	13- BN_Input_Volt_ST
	14	14- BN_Input_Volt_ST20
	15	15- BN_Input_Volt_ST80
	16	16- RN_Output_Volt_ST

Event Register F:

	1	1- RN_Output_Volt_ST20
	2	2- RN_Output_Volt_ST80
	3	3- YN_Output_Volt_ST
	4	4- YN_Output_Volt_ST20
	5	5- YN_Output_Volt_ST80
	6	6- BN_Output_Volt_ST
	7	7- BN_Output_Volt_ST20
F	8	8- BN_Output_Volt_ST80
	9	9- Supply Outage Source
	10	10- Supply Outage Load
	11	11- Rear device healthy
	12	12- Ph_Seq_Supervision
	13	13- Spare2_ST
	14	14- Spare3_ST
	15	15- Spare4_ST
	16	16- Spare5_ST

Event Register G:

	1	1- AR_Reclaim_Stater
	2	2- AR_PF_Dead1_Starter
	3	3- AR_PF_Dead2_Starter
	4	4- AR_PF_Dead3_Starter
	5	5- AR_PF_Dead4_Starter
	6	6- AR_EF_Dead1_Starter
	7	7- AR_EF_Dead2_Starter
	8	8- AR_EF_Dead3_Starter
G	9	9- AR_EF_Dead4_Starter
	10	10- AR_SEF_Dead1_Starter
	11	11- AR_SEF_Dead2_Starter
	12	12- AR_SEF_Dead3_Starter
	13	13- AR_SEF_Dead4_Starter
	14	14- AR_Unsuccess
	15	15- AR_Success

	16	16- Device_Restart

Event Register H:

H	1	1- Device Reset Local
	2	2- Device Reset Remote
	3	3- Setting Change Local
	4	4- Setting Change Remote
	5	5- Fault Record Clr Local
	6	6- Fault Record Clr Remote
	7	7- Event Record Clr Local
	8	8- Event Record Clr Remote
	9	9- Relay Test
	10	10- LED Test
	11	11- Trip Count Alarm
	12	12- I^2 Alarm
	13	13- Trip Count Alarm Local
	14	14- Trip Count Alarm Remote
	15	15- I^2 Alarm Clr Local
	16	16- I^2 Alarm Clr Remote

Event Register I:

I	1	1- PF_IDMT Block
	2	2- EF_IDMT Block
	3	3- PF Wideset Block
	4	4- EF Wideset Block
	5	5- PF Highset Block
	6	6- EF Highset Block
	7	7- OV_STG1_Block
	8	8- OV_STG2_Block
	9	9- UV_STG1_Block
	10	10- UV_STG2_Block
	11	11- Res.V_STG1_Block
	12	12- Res.V_STG2_Block
	13	13- SEF_Block
	14	14- AR Initialization
	15	15- Control Changed
	16	16- Spare2

Event Register J:

	1	1- Digital Input1 Block
	2	2- Digital Input2 Block
	3	3- Digital Input3 Block
	4	4- Digital Input4 Block

J	5	5- Digital Input5 Block
	6	6- Digital Input6 Block
	7	7- Digital Input7 Block
	8	8- Digital Input8 Block
	9	9- Digital Input9 Block
	10	10- Digital Input10 Block
	11	11- Digital Input11 Block
	12	12- Digital Input12 Block
	13	13- Digital Input13 Block
	14	14- Digital Input14 Block
	15	15- Digital Input15 Block
	16	16- Hot Line Tag

Event Register K:

	1	1- Push_Button1
	2	2- Push_Button2
	3	3- Block Close
	4	4- Block Trip
	5	5- Auto Reclose Block
	6	6- Group Change
	7	7- Manual Trip
K	8	8- Remote Trip
	9	9- Local mode/Remote mode
	10	10- Dead line Active
	11	11- CBFB ST
	12	12- CBFB Trip
	13	13- Alarm Accept
	14	14- Trip lock
	15	15- AR L/R Time
	16	16- Manual Close

Event Register L:

L	1	1- DI1_pk_dp
	2	2- DI2_pk_dp
	3	3- DI3_pk_dp
	4	4- DI4_pk_dp
	5	5- DI5_pk_dp
	6	6- DI6_pk_dp
	7	7- DI7_pk_dp
	8	8- DI8_pk_dp
	9	9- DI9_pk_dp
	10	10- DI10_pk_dp
	11	11- DI11_pk_dp

	12	12- DI12_pk_dp
	13	13- DI13_pk_dp
	14	14- DI14_pk_dp
	15	15- DI15_pk_dp
	16	16- PB1_pk_dp

Event Register M:

M	1	1- PB2_pk_dp
	2	2- Device Healthy
	3	3- DC Healthy
	4	4- TCS
	5	5- Breaker Status
	6	6- Protection Healthy/UnHealthy
	7	7- Spring Charge
	8	8- Cold Load Raise
	9	9- Cold Load Timer ON/OFF
	10	10- Device Restart
	11	11- EEPROM Error
	12	12- Outage Bus Clear
	13	13- Outage Line Clear
	14	14- Spare3
	15	15- KVA Demand Clear
	16	16- Remote Close

For further information contact:

MEGAWIN SWITCHGEAR PRIVATE LIMITED

Post Box No. 434,
Perumal Malai Adivaram, Alagapuram,
Salem - 636 004, Tamilnadu, India.

Phone : **91- 427- 2447099, 2430571**
E – Mail : **relay@megawinswitchgea**
Web : **www.megawinswitchgear.**

13/07/2018
28/07/2018
31/7/2018
19/11/2018
19/12/2019
16/11/2020
19/03/2022

……6
……8
……..10

…….11

…….12

…….12

…….12

…….12

…….14

neters. The following parameters

lication	**Access**
.001	R
.001	R

.001	R
.001	R
.001	R
.001	R
.001	R
.001	R
.001	R
.001	R
.001	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
)01	R
.001	R
.001	R
.001	R

	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R

	R
	R
	R

	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R
	R

on	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R
	R
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

tion	Access
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

SEF Set
1-120 A
5-500 A
15-1500 A

	RES Stage2
	20-600V
	40-1200V
	60-2000V
	120-4000V
	180-6000V

ı for the other groups are
all the Groups. The starting

on	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R_W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

n	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W

ion	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

n	Access
	R-W
	R-W

n	Access
	R-W
	R-W

	R-W
	R-W

plication actor	Access
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W

1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W
1	R-W

ation	Access
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

for the same Setting

Byte we can set all the eight

ation or	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

on	Access
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W

	R-W
	R-W
	R-W

	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W
	R-W

	Access
	R
	R
	R
	R
	R
	R
	R
	R

	Access
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W
	W

Access
W
W
W
W
W
W

n	Access
	R
	R
	R

	R

Access
R
R
R
R

R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R

Access
R
R
R
R
R
R
R
R
R

R
R
R
R

., 2430572.
ur.com
.com

Nov-2020

Address	Register Name	Range	Register	Step	Multiplication Factor
40017	R Phase Meter Current	-	4 Bytes	-	0.001
40019	Y Phase Meter Current	-	4 Bytes	-	0.001
40021	B Phase Meter Current	-	4 Bytes	-	0.001
40023	R Phase L-N Voltage	-	4 Bytes	-	0.0001
40025	Y Phase L-N Voltage	-	4 Bytes	-	0.0001
40027	B Phase L-N Voltage	-	4 Bytes	-	0.0001
40029	RY Line-Line Voltage	-	4 Bytes	-	0.0001
40031	YB Line-Line Voltage	-	4 Bytes	-	0.0001
40033	BR Line-Line Voltage	-	4 Bytes	-	0.0001
40049	R Phase Reverse KW	-	4 Bytes	-	0.001
40051	R Phase Forward KW	-	4 Bytes	-	0.001
40053	Y Phase Reverse KW	-	4 Bytes	-	0.001
40055	Y Phase Forward KW	-	4 Bytes	-	0.001
40057	B Phase Reverse KW	-	4 Bytes	-	0.001
40059	B Phase Forward KW	-	4 Bytes	-	0.001
40073	R Phase Power Factor	-	4 Bytes	-	0.001
40075	Y Phase Power Factor	-	4 Bytes	-	0.001
40077	B Phase Power Factor	-	4 Bytes	-	0.001
40103	Total KW Reverse	-	4Bytes	-	0.001
40105	Total KW Forward	-	4 Bytes	-	0.001
40107	Total KVA Reverse	-	4 Bytes	-	0.001
40109	Total KVA Forward	-	4 Bytes	-	0.001
40111	Total KVAR Reverse Lag	-	4 Bytes	-	0.001
40113	Total KVAR Reverse Lead	-	4 Bytes	-	0.001
40115	Total KVAR Forward Lag	-	4 Bytes	-	0.001
40117	Total KVAR Forward Lead	-	4 Bytes	-	0.001
40119	R Active Energy Reverse	-	4 Bytes	-	0.001
40121	Y Active Energy Reverse	-	4 Bytes	-	0.001
40123	B Active Energy Reverse	-	4 Bytes	-	0.001
40125	R Active Energy Forward	-	4 Bytes	-	0.001
40127	Y Active Energy Forward	-	4 Bytes	-	0.001
40129	B Active Energy Forward	-	4 Bytes	-	0.001
40167	Total Active Energy Reverse	-	4 Bytes	-	0.001
40169	Total Active Energy Forward	-	4 Bytes	-	0.001
40171	Total Apparent Energy Reverse	-	4 Bytes	-	0.001
40173	Total Apparent Energy Forward	-	4 Bytes	-	0.001
40175	Total KVARH Reverse Lead	-	4 Bytes	-	0.001
40177	Total KVARH Reverse Lag	-	4 Bytes	-	0.001
40179	Total KVARH Forward Lead	-	4 Bytes	-	0.001
40181	Total KVARH Forward Lag	-	4 Bytes	-	0.001
40183	Average Power Factor	-	4 Bytes	-	0.001
40201	Frequency	-	4 Bytes	-	0.01
40215	Auto Trip Count	-	4 Bytes	-	1
40061	R Phase Reverse KVA	-	4 Bytes	-	0.001
40063	R Phase Forward KVA	-	4 Bytes	-	0.001

40065	Y Phase Reverse KVA	-	4 Bytes	-	0.001
40067	Y Phase Forward KVA	-	4 Bytes	-	0.001
40069	B Phase Reverse KVA	-	4 Bytes	-	0.001
40071	B Phase Forward KVA	-	4 Bytes	-	0.001
40001	R Phase Protection Current	-	4 Bytes	-	0.001
40003	Y Phase Protection Current	-	4 Bytes	-	0.001
40005	B Phase Protection Current	-	4 Bytes	-	0.001
40007	Earth Current	-	4 Bytes	-	0.001
40009	Sensitive Earth Fault Current	-	4 Bytes	-	0.001
40011	I1 Current-Positive Sequence	-	4 Bytes	-	0.001
40013	I2 Current-Negative Sequence	-	4 Bytes	-	0.001
40015	I0 Current-Zero Sequence	-	4 Bytes	-	0.001
40035	Residual Voltage	-	4 Bytes	-	0.0001
40037	V1 Voltage-Positive Sequence	-	4 Bytes	-	0.0001
40039	V2 Voltage-Negative Sequence	-	4 Bytes	-	0.0001
40041	V0 Voltage-Zero Sequence	-	4 Bytes	-	0.0001
40043	R Phase L-N Output Voltage	-	4 Bytes	-	0.0001
40045	Y Phase L-N Output Voltage	-	4 Bytes	-	0.0001
40047	B Phase L-N Output Voltage	-	4 Bytes	-	0.0001
40079	R Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40081	R Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40083	R Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40085	R Phase KVAR Forward Lead	-	4 Bytes	-	0.001
40087	Y Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40089	Y Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40091	Y Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40093	Y Phase KVAR Forward Lead	-	4 Bytes	-	0.001
40095	B Phase KVAR Reverse Lag	-	4 Bytes	-	0.001
40097	B Phase KVAR Reverse Lead	-	4 Bytes	-	0.001
40099	B Phase KVAR Forward Lag	-	4 Bytes	-	0.001
40101	B phase KVAR Forward Lead	-	4 Bytes	-	0.001
40131	R Apparent Energy Reverse	-	4 Bytes	-	0.001
40133	Y Apparent Energy Reverse	-	4 Bytes	-	0.001
40135	B Apparent Energy Reverse	-	4 Bytes	-	0.001
40137	R Apparent Energy Forward	-	4 Bytes	-	0.001
40139	Y Apparent Energy Forward	-	4 Bytes	-	0.001
40141	B Apparent Energy Forward	-	4 Bytes	-	0.001
40143	R KVARH Reverse Lead	-	4 Bytes	-	0.001
40145	Y KVARH Reverse Lead	-	4 Bytes	-	0.001
40147	B KVARH Reverse Lead	-	4 Bytes	-	0.001
40149	R KVARH Reverse Lag	-	4 Bytes	-	0.001
40151	Y KVARH Reverse Lag	-	4 Bytes	-	0.001
40153	B KVARH Reverse Lag	-	4 Bytes	-	0.001
40155	R KVARH Forward Lead	-	4 Bytes	-	0.001
40157	Y KVARH Forward Lead	-	4 Bytes	-	0.001
40159	B KVARH Forward Lead	-	4 Bytes	-	0.001
40161	R KVARH Forward Lag	-	4 Bytes	-	0.001
40163	Y KVARH Forward Lag	-	4 Bytes	-	0.001
40165	B KVARH Forward Lag	-	4 Bytes	-	0.001
40185	MD Apparent Power Reverse	-	4 Bytes	-	0.001

40187	MD Apparent Power Forward	-	4 Bytes	-	0.001
40189	MD Active Power Reverse	-	4 Bytes	-	0.001
40191	MD Active Power Forward	-	4 Bytes	-	0.001
40193	Cumulative MD KVA Reverse	-	4 Bytes	-	0.001
40195	Cumulative MD KVA Forward	-	4 Bytes	-	0.001
40197	Cumulative MD KW Reverse	-	4 Bytes	-	0.001
40199	Cumulative MD KW Forward	-	4 Bytes	-	0.001
40203	Last Outage Bus Days-2 Bytes,Hours,Minutes	-	4Bytes	-	1
40205	Total Outage Bus Days-2 Bytes,Hours,Minutes	-	4 Bytes	-	1
40207	Last Outage Load Days-2 Bytes,Hours,Minutes	-	4Bytes	-	1
40209	Total Outage Load Days-2 Bytes,Hours,Minutes	-	4 Bytes	-	1
40211	Outage Count Bus	-	4 Bytes	-	1
40213	Outage Count Load	-	4 Bytes	-	1
40217	Sum Of I^2 Alarm	-	4 Bytes	-	1
25024	3Ph Total KW	-	4 Bytes	-	0.001
25026	3Ph Total KVAR	-	4 Bytes	-	0.001
25028	3Ph Total KVA	-	4 Bytes	-	0.001

Access
R
R
R

R
R
R

R
R
R
R
R
R
R
R
R
R
R
R
R
R
R
R